# Variables

*Which variables will you include? How will these variables be coded — will you use them the way they are or recode them, and if so, how?*

## Outcomes

**H1:** “Best if man and woman work the same amount so they can share…” (plh0308_v2). Coded as is, from 1 to 7, with higher values indicating stronger agreement (more egalitarian attitudes).

**H2**: Political attitude, left-right (plh0004). Reverse-coded (10-x), so greater values indicate more left-leaning political preferences.

**H3:** Difference in weekly unpaid work hours between father and female partner (which need not be the mother of the father’s children), calculated as the father’s hours minus the partner’s hours, so higher values mean greater egalitarianism. Unpaid work hours are calculated as the sum of reported average hours spent on housework, errands/shopping, and repairs.

Weekly hours for each partner are calculated as five times reported average housework/errands/repairs hours on weekdays (pli0043_v3/pli0040/pli0049_v3) + reported average housework/repairs hours on Saturdays (pli0012_v3/???/pli0031_v4) + reported average housework/repairs hours on Sundays (pli0016_v4/pli0011/pli0034_v4), with the “???” indicating that we were not able to identify the long-format variable name for hours on errands on Saturdays based on SOEP documentation, but inspection of a few questionnaires suggests that this variable should be available.

In a full analysis, we would conduct supplementary analyses considering hours spent on childcare (separately and as a further component of total unpaid work), but here we want to restrict the analysis to one outcome variable per hypothesis to reach an unambiguous conclusion on whether the hypothesis is supported.

**H4:** Same as H1.

## Covariates

The following variables will be included in all regressions, as further described below. We expect child gender to be as good as exogenous but nevertheless include a few

covariates to increase precision and because there may be some selection into further higher-parity births based on the sex/gender composition of previous births.

- Indicator variable for having at least one (biological) daughter ("AnyDaughter"): at least one female child (1) *vs*. only (biological) male children (0). We expect to construct this variable using the variable sex in the biobirth file as well as the variables kidgeb?? to consider only children that were born before a given survey year. If possible, we will further refine this using information on the month of the interview and children's months of birth. The restriction to (biological) is not primarily based on practical considerations, since the data in the biobirth file refer to likely biological children only.
- Number of (biological) children, included using a full set of dummies/indicator variables. Constructed using information in biobirth data, number of rows (= children) for adult respondent, only counting those who were born before a given survey year/interview (see previous variable).
- Respondent age (quinquennial groups 18-24, 25-29, 30-34 etc.), constructed by subtracting gebjahr from the variable syear. Included in the regressions as a full set of dummies/indicator variables.
- Education (pgisced97(: ISCED-97 levels 0-2 (low); levels 3-4 (medium); levels 5-6 (high), full set of dummies/indicator variables.
- Age at first birth (quinquennial groups 18-24, 25-29, 30-34 etc.), full set of dummies/indicator variables.
- Religion (plh0258_h): categorical (Christian; Islamic; Other), full set of dummies/indicator variables.
- H2-4: + Survey year dummies.

# Inclusion/exclusion criteria

*Are there any observations you will exclude? If yes, which? Which years will you include/what period will you study?*

We restrict the analyses of Hypotheses 1-3 to fathers, because we believe a potential "daughter effect" to be much more likely/theoretically plausible to this group (parts of the prior literature seem to have done so as well). We will only consider mothers in testing H4, as it is explicitly about a gender difference in the daughter effect. Childless respondents are excluded from all analyses.

Specifically, we will apply the following inclusion criteria restrictions:

- H1-2: Include all fathers, i.e., adult (18+) respondents with at least one child who self-identify as male (irrespective of partnership status, co-residence with children or sexual orientation)
- For H3: Include all fathers living with female partners (irrespective of co-residence with children or sexual orientation)

- H4: Include all male or female parents, i.e., adult respondents with at least one child who self-identify as male or female (irrespective of partnership status, co-residence with children or sexual orientation)
- We exclude parents who have any children with non-binary gender as these cases are ambiguous with respect to the focal treatment.

## Survey waves

We will generally use all survey waves that include all measures used in the analysis for a given hypothesis. While information on the independent variables should be available for all survey waves (subject to item non-response/missing data), inspection of the SOEP documentation indicates that the outcome measures were included only in the following years:

- H1/H4: 2018
- H2: 2005, 2009, 2015, 2019
- H3: 2001-2019 (odd years)

We note that we will use any additionally available years that the outcome measures are available for, if inspection of the SOEP data (which we do not yet have access to) reveals this to be the case.

# Exact analysis

*Please write down the exact specification of the regression model, including the names of the dependent and independent variables, and whether you use robust standard errors or something else.*

## Regression models

We will estimate linear regressions on pooled cross-sectional samples for all hypotheses/outcomes, including only observations with valid data on all predictors and the respective outcome variable (complete case analysis). We will cluster standard errors at the individual level for H1-H3 and at the household level for H4, using the default clustering correction in Stata, as implemented in `vce(cluster clustervar)`. Throughout, we will use weighted least squares using the cross-sectional person weight variable (`phrf`).

We will use the same regression specification to test hypotheses H1-3 (the three outcome variables listed above). All covariates (listed above), including the total number of children will be included as categorical predictors (i.e. using a full set of dummy variables). To test H4, we will use the same outcome as for H1, but will instead estimate separate models for fathers and mothers, with the assessment of the hypothesis conducted using the postestimation command `suest` (as described below).

## Interpretation of results

*Please specify which statistic you will take into account for testing the hypothesis. How do you interpret whether your test result is in line with the hypothesis or not?*

For H1-H3, we will consider the estimate of the “AnyDaughter” coefficient and consider the hypothesis supported if, a), the coefficient estimate is positively signed and, b), has a two-tailed p-value < .05.

For H4, we will use the postestimation command `suest` to combine the regression estimates for fathers and mothers. We will consider the hypothesis supported if the estimate of the “AnyDaughter” coefficient for fathers is, a), positively signed, b), larger than the corresponding estimate for mothers, and c), a test for the equality of the two coefficient estimates yields a two-tailed p-value < .05.